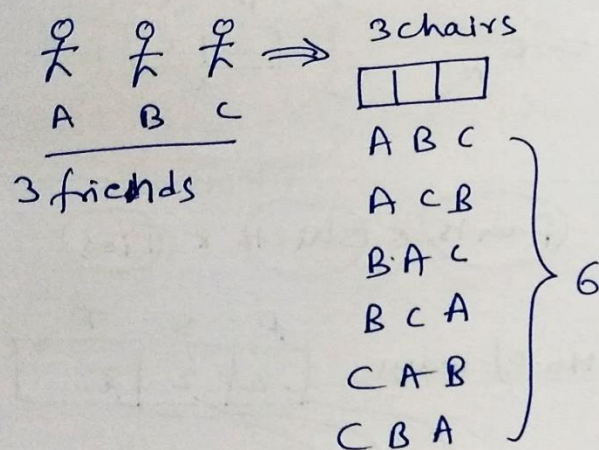


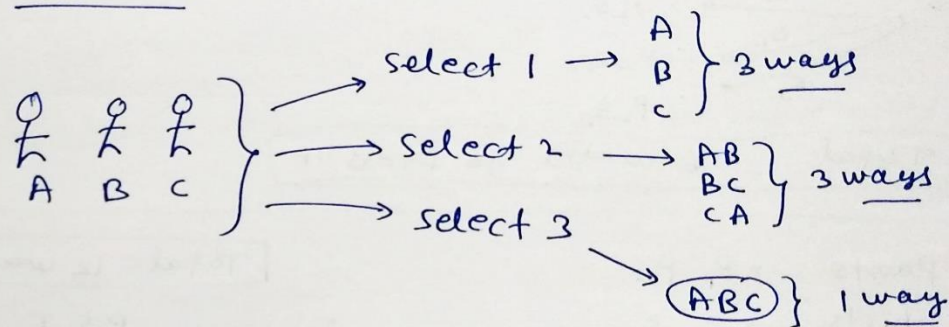
Permutations and Combinations

No. of Arrangements



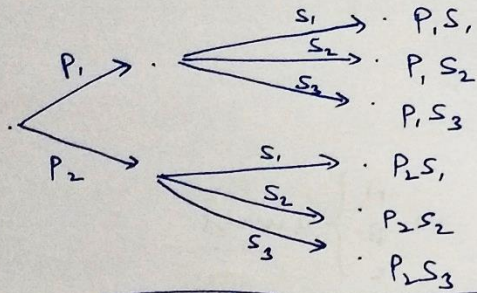
Discussed in Exercise
7.1, 7.2, 7.3

No. of Selections



Discussed in Exercise 7.4

e.g. ① 2 pants $\rightarrow P_1, P_2$
 3 shirts $\rightarrow S_1, S_2, S_3$



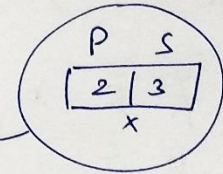
Total = 6 ways to Dress up

Solution by
Fundamental Principle of Counting
 (multiplication rule)

$$= \text{Pants} \times \text{Shirts}$$

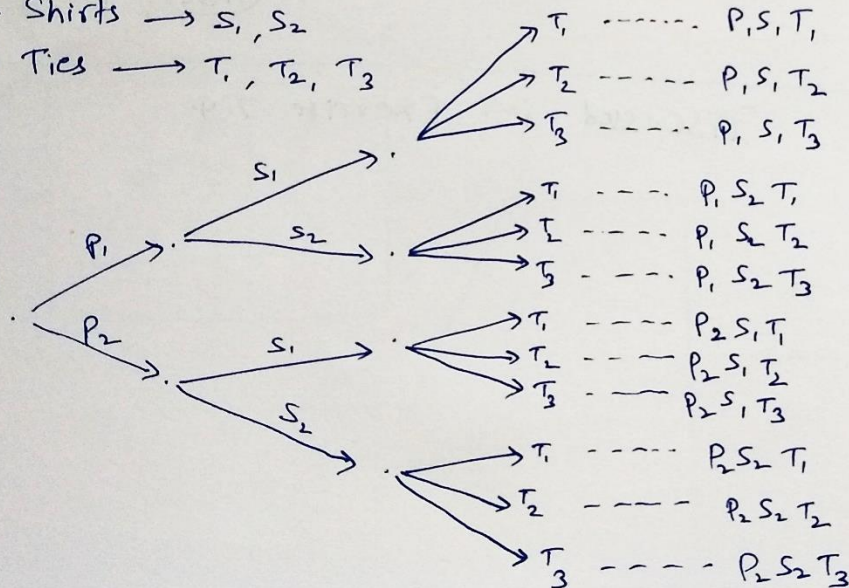
$$= 2 \times 3$$

$$= 6$$

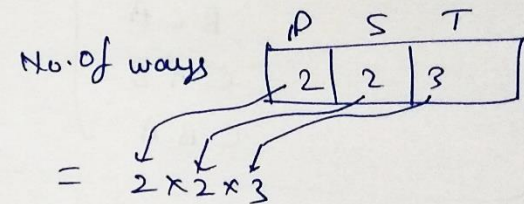


e.g. ② 2 pants $\rightarrow P_1, P_2$
 2 Shirts $\rightarrow S_1, S_2$
 3 Ties $\rightarrow T_1, T_2, T_3$

Total = 12 ways



Pants $\times$ Shirts $\times$ Ties



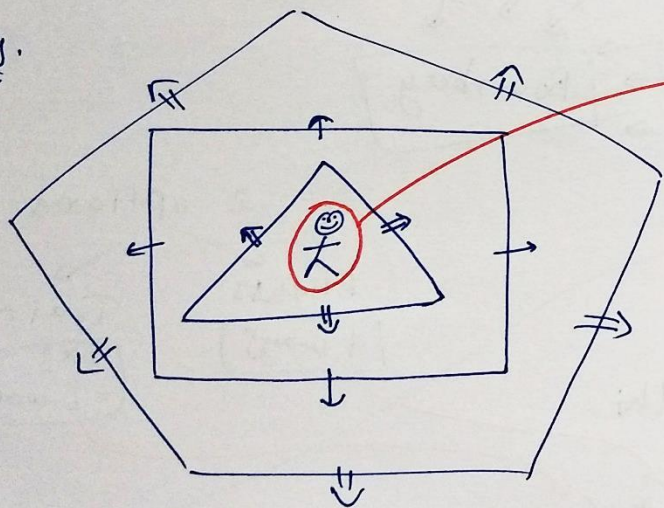
$$= 12$$

Fundamental Principle of Counting:

↓
Theorem

① Multiplication Principle. Let suppose a process is to be completed in 3 steps (I followed by II followed by III), and no. of ways to complete step I is 'm', step II $\rightarrow$ 'n', step III $\rightarrow$ 'l', then no. of ways to complete the whole process is given by $= 'm \times n \times l'$.

e.g.



No. of ways to come out of this structure = ?

$$\triangle \rightarrow 3$$

$$\square \rightarrow 4$$

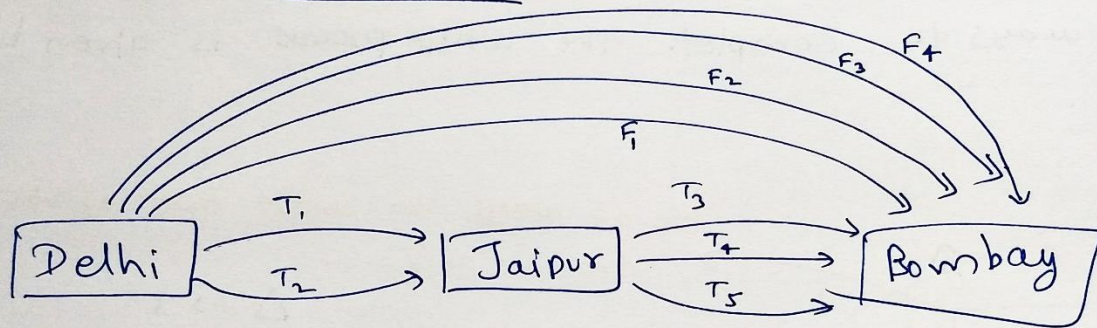
$$\pentagon \rightarrow 5$$

Total No. of in which  can come out = $3 \times 4 \times 5$

$$= 60 \quad \checkmark$$

II Addition Principle : $\rightarrow$ If a process is to be completed by 3 options. (each option can complete the whole process)
 let No. of ways to complete I option is m , $\text{II} \rightarrow n$, $\text{III} \rightarrow l$,
 then no. of ways in which the process can be completed = $m+n+l$.

e.g.



No. of ways to reach Bombay from Delhi

$$= 4 + 6 = 10 \text{ ways}$$

2 options

Flights

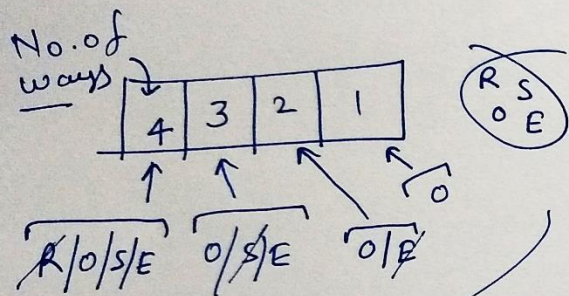
4 ways

Trains

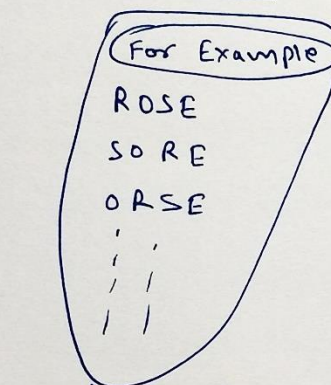
2×3
= 6 ways

e.g. No. of Different words (4 lettered) (with or without meaning)
by the letters of word "ROSE" = ?

(i) Without Repetition of Letters



R S E O

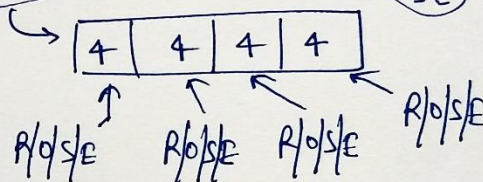


multiplication
rule

$$= 4 \times 3 \times 2 \times 1$$
$$= 24 \text{ ways}$$

(ii) with repetition of letters

No. of ways



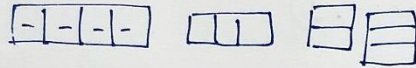
Because
Repetition
is allowed

$$= 4 \times 4 \times 4 \times 4$$

$$= 256$$

Exercise 6.1

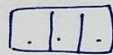
- ★ {
1. Structure
 2. Restrictions
 3. Repetition



$m+n$

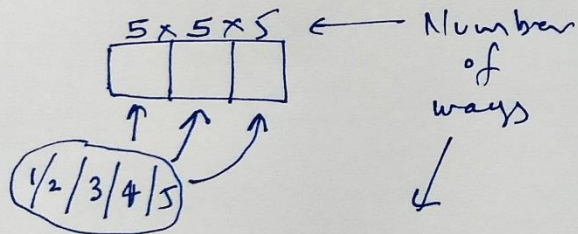
Q.1

3 Digit Numbers



Available No. = 1, 2, 3, 4, 5

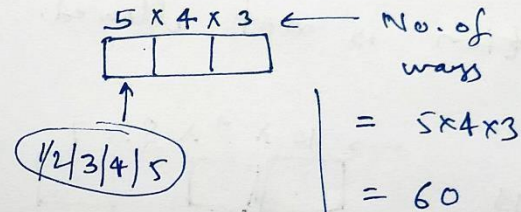
(i) Repetition Allowed.



$$= 5 \times 5 \times 5$$

$$= 125$$

(ii) Repetition is not allowed



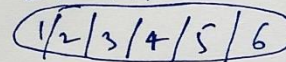
$$= 5 \times 4 \times 3$$

$$= 60$$

Q.2

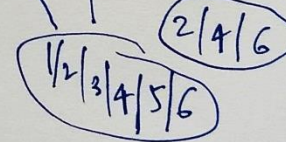
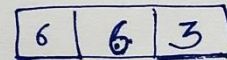
3 digit No. ✓

Even No. ✓



Repetition is Allowed

No. of ways →



$$6 \times 6 \times 3$$

$$= 108 \text{ ways.}$$

③ → "4 letter Code"

→ By First 10 letters of English Alphabet.

→ Repitition is not allowed.

↓
(a, b, c, ..., j)

No. of
ways

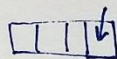
$$\begin{array}{c} \rightarrow 10 \times 9 \times 8 \times 7 \\ \begin{array}{|c|c|c|c|} \hline . & . & & \\ \hline \end{array} \\ \uparrow \\ \textcircled{a, b, \dots, j} \end{array} = 720 \times 7 = \underline{5040 \text{ ways}}$$

Q.4

① Structure

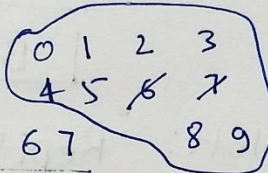
② Restrictions

③ Repetition

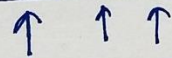
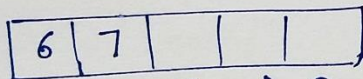


✓ 5 Digit Telephone No.

Using '0' to '9' →



✓ Each No. starts with 67
Repetition → Not Allowed.



$8 \times 7 \times 6$ ← No. of ways

$$\begin{aligned}\text{Total No. of ways} &= 8 \times 7 \times 6 \\ &= 56 \times 6 \\ &= 336 \checkmark\end{aligned}$$

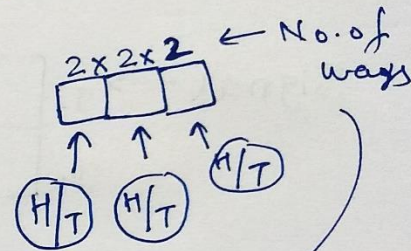
Q.5

A coin is tossed 3 times



Outcomes are recorded

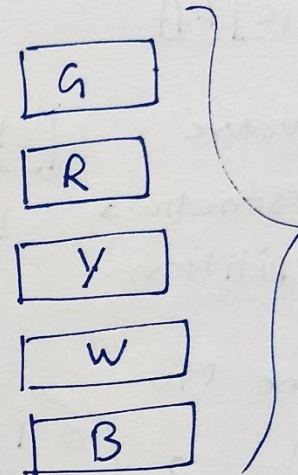
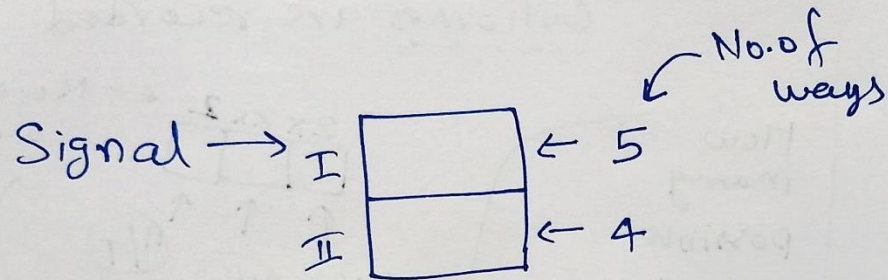
How many possible outcomes are there?



$$\begin{aligned}&= 2 \times 2 \times 2 \\ &= 8\end{aligned}$$

[Q.6]

5 flags of Different Colours →



Total No. of Signals

$$= 5 \times 4$$

$$= 20 \checkmark$$

Note: "Repetition is not" Allowed

Self understood

Factorial x Exercise 6.2

$$\text{Factorial of '5'} = 5 \times 4 \times 3 \times 2 \times 1 = 5! = \underline{5}$$

$$\text{Factorial of '7'} = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 7! = \underline{7}$$

$$n! = \underline{n}$$

Definition of factorial of 'n' ($n \in \mathbb{N}$)

= the product of first n natural numbers.

$$= \underline{n \cdot (n-1) \cdot (n-2) \cdot (n-3) \cdot \dots \cdot (2)(1)}$$

Note

$$\boxed{\text{Factorial of '0'} = 0! = \underline{0} = 1}$$

$$\rightarrow 0! = 1$$

$$N \left\{ \begin{array}{l} 1! = 1 \\ 2! = 2 \times 1 = 2 \\ 3! = 3 \times 2 \times 1 = 6 \end{array} \right.$$

$(-2)! \times$
 $(\frac{1}{2})! \times$ → Not defined

e.g.

$$10! = \underline{10}$$

Diagram illustrating the recursive calculation of $9!$ using three horizontal lines:

- The top line contains the sequence of numbers from 10 down to 1.
- The middle line shows arrows indicating the progression of the factorial calculation:
 - An arrow labeled $9!$ spans from 10 to 9.
 - An arrow labeled $8!$ spans from 9 to 8.
 - An arrow labeled $6!$ spans from 7 to 6.
- A vertical dashed line is drawn between 6 and 7 on the top line.

$10! = 10 \times 9 \times \boxed{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$

$$10! = \underline{10 \times 9 \times 8!}$$

Property of Factorial.

$$\begin{aligned} n! &= n \times (n-1)! \\ &= n \times (n-1) \times (n-2)! \\ &= n \times (n-1) \times (n-2) \times (n-3)! \\ &\vdots \\ &= \underline{n (n-1) (n-2) (n-3) \dots 3 \cdot 2 \cdot 1} \end{aligned}$$

$$\begin{aligned} & \frac{8!}{4!} \\ &= \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{4 \cdot 3 \cdot 2 \cdot 1} \\ &= 8 \times 7 \times 6 \times 5 \\ &= 1680 \end{aligned}$$

Don't's

$$8! + 7! \neq (8+7)! = 15! \quad \times$$

$$\frac{8!}{4!} = 2!$$

Q.1

$$(i) 8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$$

$$(ii) 4! - 3! = 4 \times 3 \times 2 \times 1 - 3 \times 2 \times 1 \\ = 24 - 6 = 18 \checkmark$$

Q.2

~~Is~~ Is $\underline{3! + 4! = 7!}$? $\longrightarrow$ wrong

$$\text{LHS} = 3! + 4! = \underline{3 \times 2 \times 1} + \underline{4 \times 3 \times 2 \times 1} = 6 + 24 = 30$$

$$\text{RHS} = 7! = \underline{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} = 5040$$

Q.3

Compute.

$$\frac{8!}{6! \times (2!)} = \frac{\cancel{8} \times 7 \times \cancel{6!}}{\cancel{6!} \times \cancel{(2 \times 1)}} = 28 \checkmark$$

④ If $\frac{1}{6!} + \frac{1}{\cancel{7}!} = \frac{n}{8!}$, find (n) .

$$\Rightarrow \frac{1}{6!} + \frac{1}{7!} = \frac{n}{\textcircled{8!}}$$

$$\Rightarrow \frac{8!}{6!} + \frac{8!}{7!} = \textcircled{n}$$

$$\Rightarrow n = \frac{8 \times \cancel{7} \times \cancel{6!}}{\cancel{6!}} + \frac{8 \times \cancel{7!}}{\cancel{7!}} = 56 + 8 = 64 \checkmark$$

⑤ Evaluate $\frac{n!}{(n-r)!}$

(i) $n=6, r=2$

$$\begin{aligned} \frac{n!}{(n-r)!} &= \frac{6!}{(6-2)!} \\ &= \frac{6!}{4!} = \frac{6 \cdot \cancel{5} \cdot \cancel{4!}}{\cancel{4!}} = 30 \end{aligned}$$

(ii) $n=9, r=5$

$$\begin{aligned} \frac{n!}{(n-r)!} &= \frac{9!}{(9-5)!} = \frac{9!}{4!} \\ &= \frac{\boxed{9 \times 8 \times 7 \times 6 \times 5} \times \cancel{4!}}{\cancel{4!}} \\ &= 15120 \checkmark \end{aligned}$$

Theory Before Ex. 6.3

Permutations

Arrangements

Example (16)

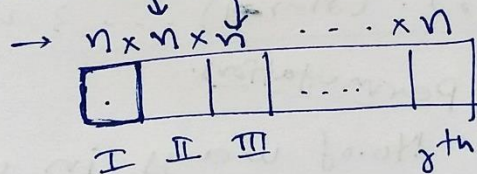
Case-I No. of Arrangements of n -objects

(all are different) at r -places

(repetition is Allowed).

CAMP

No. of
ways



$$= n \times n \times n \times \dots \times n$$

(r -times)

$$= n^r$$

e.g.

CAMP

Repetition
is Allowed

$$4.4.4.4 = 4 \times 4 \times 4 \times 4 \leftarrow \text{No. of ways} = 4^4 = 256$$

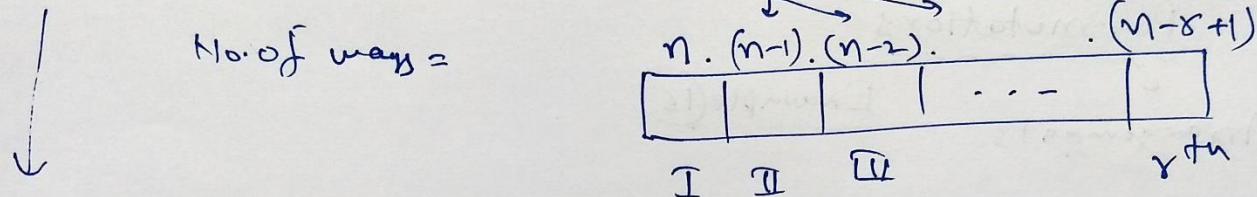
4 places

4.4.4

3 places

$$= 4 \times 4 \times 4 \leftarrow \text{No. of ways} = 4^3 = 64$$

Case-2 No. of Arrangements of n -objects (all are different) at r -places (repetition is not allowed)



$$= n \times (n-1) \times (n-2) \times \dots \times (n-r+1)$$

$$= \frac{n \times (n-1) \times (n-2) \times \dots \times (n-r+1) \cdot \boxed{(n-r) \cdot (n-r-1) \cdot \dots \cdot 3 \cdot 2 \cdot 1}}{\boxed{(n-r) \cdot (n-r-1) \cdot \dots \cdot 3 \cdot 2 \cdot 1}}$$

$$= \frac{n!}{(n-r)!} = n P_r = \text{Permutations.}$$

No. of ways in which n -different objects can be arranged on r -places without repetition.

Objects (circled) Seats (circled)

Note $n P_0 = \frac{n!}{(n-0)!} = \frac{n!}{n!} = 1$

$$n P_1 = \frac{n!}{(n-1)!} = \frac{n \times \cancel{(n-1)!}}{\cancel{(n-1)!}} = n$$

$$n P_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$$

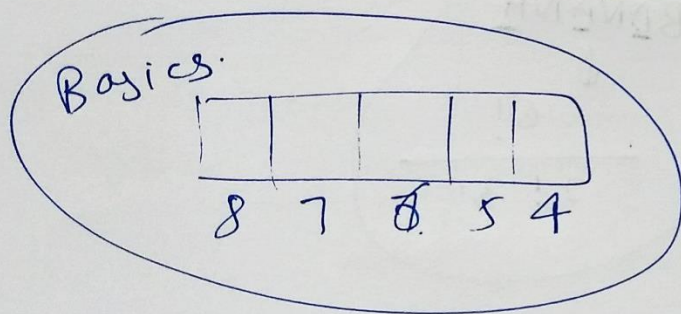
$0! = 1$ (circled)

P.g. DAUGHTER $\rightarrow$ Arrange its all letters at 8 places
(i) (without repetition)

$$= {}^8P_8 = 8! = \boxed{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

DAUGHTER $\rightarrow$ (ii) Arrange its all letters at 5 places
(without repetition)

8 letters, 5 places



Permutations

$$= {}^8P_5$$

$\leftarrow$ Available candidates
 $\leftarrow$ Places.

$$= \frac{8!}{(8-5)!} = \frac{8!}{3!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{\cancel{3 \times 2 \times 1}}$$

$$= \underline{\hspace{2cm}}$$

Case. (II) No. of Arrangements of n -objects (in which 'p' are alike of 1st kind, 'q' are alike of 2nd kind)

at n places =
$$\frac{n!}{p! q!}$$

kind = type

Reason:

$$\frac{4!}{2!}$$

same (Alike)

BOOK

$\left\{ \begin{array}{l} BO_1O_2K \\ BO_2O_1K \end{array} \right.$

BANANA $\rightarrow$ total = 6

B, NN, AAA

2

3

BANANA

↓

6!

2! 3!

Example 16

Independence

total = 12

I
NNN
DD
EEEE
P
C

③
②
④

e.g.
NNN IEEEE DDPC

BAG method

Total No. of Arrangements (i) words start with 'P'

$$= \frac{12!}{3! 2! 4!}$$

Fixed
11-Places
 $= \frac{11!}{3! 2! 4!}$

All the vowels never (iii) occur together

= total - those cases in which all vowels occur together

$$= \frac{12!}{3! 2! 4!} - \frac{8! \times 5!}{3! 2! 4!}$$

(iv) words begin with 'I' & end in 'P'

Fixed
10 Places

$$= \frac{10!}{3! 2! 4!}$$

All the vowels always (ii) occur together

Consonants	vowels
NNN DD P C	I, EEEE

Consider as a one object
I EEEE

(I) → Permute NNN DD P C

$$\frac{8!}{3! 2!}$$

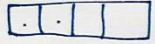
(II) In the Bag → Permute I EEEE

$$\text{Total} = \frac{8!}{3! 2!} \times \frac{5!}{4!}$$

1. Structure
2. Restriction

Basic

FIRE

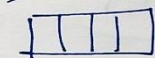


$$4 \cdot 3 \cdot 2 \cdot 1 = 4! \\ \rightarrow = 24$$



$$\frac{n!}{(n-r)!} = n P_r \rightarrow \begin{matrix} \text{Available} \\ \text{Candidates (upper} \\ \text{suffix)} \end{matrix}$$

$$\rightarrow \text{seats (lower} \\ \text{suffix)}$$



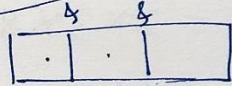
$$4 P_4 = \frac{4!}{(4-4)!} = \frac{4!}{0!} \\ = \frac{24}{1}$$

Q.1

3-digit No.

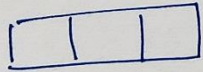
Available 1, 2, 3, 4, 5, 6, 7, 8, 9

By Basics



$$\text{No. of ways} \rightarrow 9 \times 8 \times 7 \\ = 9 \times 56 \\ = 504$$

nPr



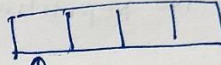
$$= 9 P_3 = \frac{9!}{6!} = \frac{9 \cdot 8 \cdot 7 \cdot 6!}{6!} \\ = 9 \times 8 \times 7 = 504$$

Q.2

4 Digit No.

?

Available
0, 1, 2, 3, 4, 5
6, 7, 8, 9,
No Repetition



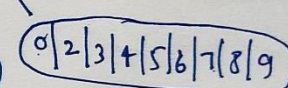
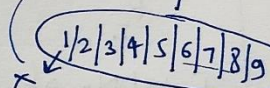
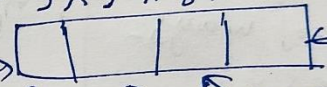
x '0'

x 05+3 $\rightarrow$ 3 digit

Indirect Restriction

No. of ways

$$\rightarrow 9 \times 9 \times 8 \times 7$$



$$\text{Total No. of ways} = 9 \times 9 \times 8 \times 7 \\ = 81 \times 56 \\ = 4536$$

$$\begin{array}{r} 81 \\ \times 56 \\ \hline 486 \\ 405 \times \\ \hline 4536 \end{array}$$

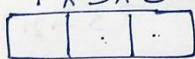
Q.3 3 digit No.

Even No.

Restriction

No. of ways

$$\rightarrow 4 \times 5 \times 3$$



2/4/6

$$\begin{aligned} \text{Total No. of ways} &= 4 \times 5 \times 3 \\ &= 60 \checkmark \end{aligned}$$

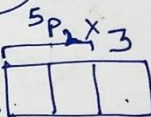
Basics.

Available

1, 2, 3, 4, 6, 7

No Repitition

nPr



nPr

2/4/6

$$= {}^5P_2$$

$$\text{Total No. of ways} = {}^5P_2 \times 3$$

$$= \frac{5!}{3!} \times 3$$

$$= \frac{5 \times 4 \times \cancel{3}}{\cancel{3!}} \times 3$$

$$= 60$$

Q.4

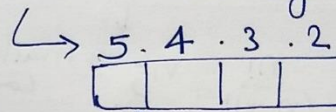
4 digit No.

Available

1, 2, 3, 4, 5

No Repitition

(i) Total No. of 4 Digit No.

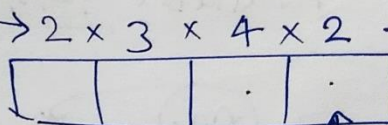


No Restriction

$$\begin{aligned} &= 5 \times 4 \times 3 \times 2 \\ &= 120 \checkmark \end{aligned}$$

(ii) Total No. of Even No. (4 Digit.)

RESTRICTION

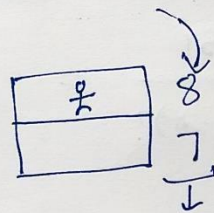


$$\begin{aligned} &= 2 \times 3 \times 4 \times 2 \\ &= 48 \end{aligned}$$

Q.5 Available $\rightarrow$ 8 persons

Chairman ?

Vice Chairman ?



and $\rightarrow$ 'x'

multiply

$$\text{Total No. of ways} = 8 \times 7 = 56$$

Q.6 Find n , if ${}^{n-1}P_3 : {}^nP_4 = 1:9$

$$\hookrightarrow {}^nP_r = \frac{n!}{(n-r)!} \Rightarrow n! = n \cdot (n-1)! = n \cdot (n-1) \cdot (n-2)! = \dots$$

$$\text{Given } \frac{{}^{n-1}P_3}{{}^nP_4} = \frac{1}{9} \Rightarrow \frac{(n-1)!}{(n-1-3)!} \cdot \frac{n!}{(n-4)!} = \frac{1}{9}$$

$$\Rightarrow \frac{(n-1)!}{(n-4)!} \cdot \frac{n!}{(n-4)!} = \frac{1}{9}$$

$$\Rightarrow \frac{(n-1)!}{n!} = \frac{1}{9}$$

$$\Rightarrow \frac{(n-1)!}{n \cdot (n-1)!} = \frac{1}{9}$$

$$\Rightarrow \frac{1}{n} = \frac{1}{9}$$

$$\Rightarrow \boxed{n=9} \checkmark$$

Q.7 Find 'x'.

$$\boxed{{}^n P_r \quad 0 \leq r \leq n} \quad \star$$

(i) $5P_x = 2 \cdot {}^6 P_{x-1}$

$$\Rightarrow \frac{5!}{(5-x)!} = 2 \cdot \frac{6!}{[6-(x-1)]!}$$

$$\Rightarrow \frac{5!}{(5-x)!} = 2 \times \frac{6!}{(7-x)!}$$

$$\Rightarrow \frac{\cancel{5!}}{(5-x)!} = 2 \times \frac{6 \times \cancel{5!}}{(7-x) \cdot (6-x) \cdot \cancel{(5-x)!}}$$

$$\Rightarrow (7-x) \cdot (6-x) = 12$$

$$\Rightarrow \cancel{4^2} - 13x + x^2 = 12$$

$$\Rightarrow x^2 - 13x + 30 = 0$$

$$\Rightarrow \underline{x^2 - 10x} - \underline{3x + 30} = 0$$

$$\Rightarrow x(x-10) - 3(x-10) = 0$$

$$\Rightarrow (x-3)(x-10) = 0$$

$$\boxed{x = 3, 10}$$

$x=3$ $\checkmark$ $5P_3 = 2 \times {}^6 P_{3-1}$ $\checkmark$
 $x=10$ $\times$ $5P_{10} = 2 \times {}^6 P_9$ $\times$
 $(10 > 5)$ $\times$

(ii) $5P_x = {}^6 P_{x-1}$

$$\Rightarrow \frac{5!}{(5-x)!} = \frac{6!}{[6-(x-1)]!}$$

$$\Rightarrow \frac{\cancel{5!}}{(5-x)!} = \frac{6 \times \cancel{5!}}{(7-x) \cdot (6-x) \cdot \cancel{(5-x)!}}$$

$$\Rightarrow (7-x)(6-x) = 6$$

$$\Rightarrow 42 - 13x + x^2 = 6$$

$$\Rightarrow x^2 - 13x + 36 = 0$$

$$\Rightarrow \underline{x^2 - 9x} - \underline{4x + 36} = 0$$

$$\Rightarrow x(x-9) - 4(x-9) = 0$$

$$\Rightarrow (x-4)(x-9) = 0$$

$$\boxed{x = 4, 9}$$

$x=4$ $\checkmark$
 $5P_4 = {}^6 P_{4-1}$ $\checkmark$

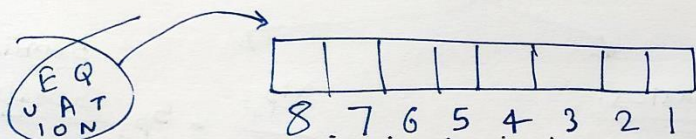
$x=9$ $\times$
 $5P_9 = {}^6 P_{9-1}$ $\times$

Q.8 EQUATION

Available $\rightarrow$ All the letters.

Repetition $\rightarrow$ No.

$${}_nP_r = \frac{n!}{(n-r)!}$$



$$\begin{aligned} \text{By Basics} &= 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 8 \times 5040 \\ &= 40320 \checkmark \end{aligned}$$

$$\text{By } {}_nP_r \begin{matrix} \leftarrow \text{Available candidates} \\ \leftarrow \text{No. of seats} \end{matrix} = \frac{n!}{(n-r)!}$$

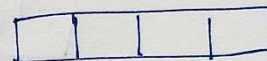
$$= {}_8P_8 = \frac{8!}{(8-8)!} = \frac{8!}{0!} = \frac{40320}{1} \checkmark$$

Q.9

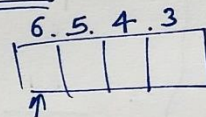
MONDAY

Repetition
 $\downarrow$
Not Allowed

(i) 4 letters at a time.



Basics.



M|O|N|D|A|Y

Total No. of words = 6.5.4.3

$$= 360.$$



$\xrightarrow{\text{Available}} {}_nP_r \rightarrow \text{Seats}$

$$= {}_6P_4$$

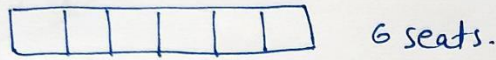
$$= \frac{6!}{2!}$$

$$= \frac{720}{2}$$

$$= 360 \checkmark$$

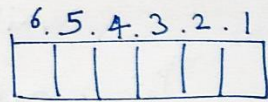
MONDAY

(ii) all letters are used at a time.



6 seats.

Basic

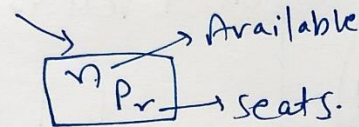


M|O|N|D|A|Y

Total No. of

$$\text{words} = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$= 720$$



$$= {}^6P_6$$

$$= \frac{6!}{(6-6)!}$$

$$= \frac{6!}{0!} = \frac{720}{1}$$

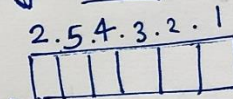
(iii) all letters are used but first letter is a vowel.

MONDAY → Vowels (O, A)
→ (M, N, D, Y)

Restriction ← Priority.



Basics.



⓪/A
A|M|N|D|Y

No. of words

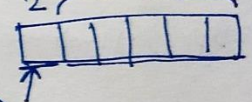
$$= 2 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 2 \times 120$$

$$= \underline{240}$$

nPr

$= 5P_5$



⓪/A

No. of words

$$= 2 \times 5P_5$$

$$= 2 \times \frac{5!}{0!}$$

$$= 2 \times \frac{120}{1} = \underline{240}$$

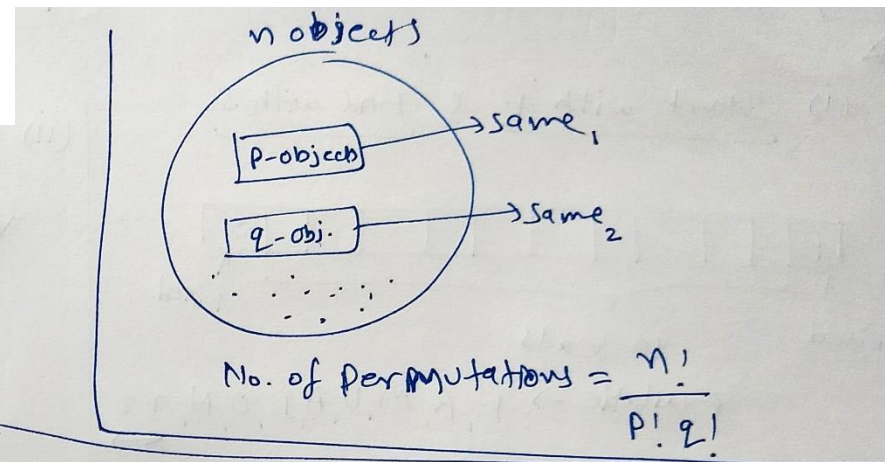
* Q.10

MISSISSIPPI $\rightarrow$ M SSSS PP
IIII

No. of words
in which
all I's do not
occur together
?

= Total No. of words - No. of words in which all I's occur together.

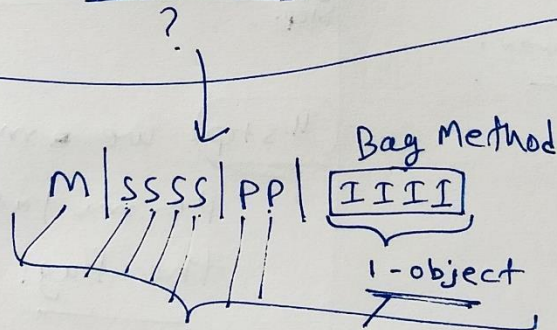
$$= \frac{11!}{4! 2! 4!} - \frac{8!}{4! 2!} = \underline{\underline{\text{Answer.}}}$$



M | SSSS | PP | IIII $\rightarrow$ ①

= $\frac{11!}{4! 2! 4!}$ Total.

SSSS $\rightarrow$ 4! PP $\rightarrow$ 2! IIII $\rightarrow$ 4!



= $\frac{8!}{4! 2!}$

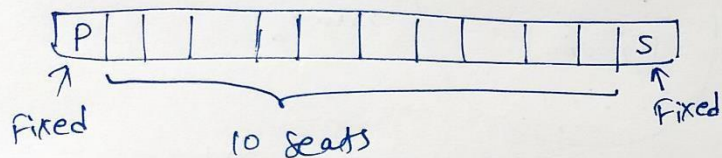
SSSS $\rightarrow$ 4! PP $\rightarrow$ 2!

★ Example 16

Q.11 PERMUTATIONS $\rightarrow$ total = 12

★

(i) Start with 'P' & End with 'S'



Available $\rightarrow$ E, R, M, U, A, I, O, N, T, T

$$\text{No. of permutations} = \frac{10!}{2!}$$

$$= \frac{10!}{2!} = \underline{\underline{\text{Answer}}}$$

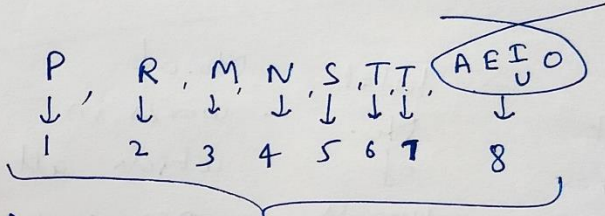
P | E | R | M | U | A | I | O | N | S | T | T

(ii) Vowels are together

BAG METHOD

~~Vowels~~ Vowels $\rightarrow$ E, U, A, I, O

Consonants $\rightarrow$ P, R, M, N, S, T, T



I Arrange these only, = $\frac{8!}{2!} \leftarrow \text{TT}$

II step we must also consider the permutations of AEIOU inside the Bag. = $\underline{\underline{5!}}$

Total No. of words in which all vowels occur together

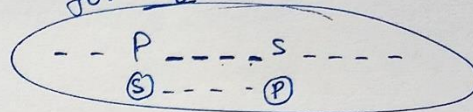
$$= \frac{8!}{2!} \times 5! = \underline{\underline{\text{Answer}}}$$

(iii) "PERMUTATIONS"

↳ there are always 4 letters between P & S?

~~P, S~~, E, R, M, V, A, I, O, N, T, T
 10
 same

for e.g:



2 { P ————— S
 S ————— P

Similarity

X	.	.	.	.	X	.	.	.	.	.	.
.	X	.	.	.	.	X	.	.	.	.	.
.	.	X	.	.	.	.	X	.	.	.	.
.	.	.	X	.	.	.	.	X	.	.	.
.	.	.	.	X	.	.	.	.	X	.	.
.	.	.	.	.	X	.	.	.	.	X	.
.	.	.	.	.	.	X	.	.	.	.	X

{ P ——— S
 S ——— P }

Remaining seats 10

$$2 \times \frac{10!}{2!} \rightarrow \text{TT}$$

Total No. of ways

$$= 7 \times 2 \times \frac{10!}{2!} = \text{Answer}$$

...X...X...

This pattern can shift 7 times.

{ P ——— S
 S ——— P }

Remaining

Ex
6.4

COMBINATIONS	${}^n C_r$	EXAMPLE - 19
--------------	------------	--------------

Selection

Permutations: No. of permutations of n -objects at ' r ' places:
(arrangement)

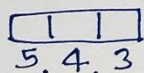
$$= {}^n P_r = \frac{n!}{(n-r)!} \quad (0 \leq r \leq n)$$

Combinations: No. of combinations of r -objects out of n -objects.
(Selection)

$$= {}^n C_r = ? \quad (0 \leq r \leq n) = \frac{n!}{(n-r)! r!}$$

e.g. 5-Persons $\rightarrow$ 3 seats (Arrange) $\rightarrow$ permutations

(i)

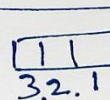


No. of Permutations = $5 \cdot 4 \cdot 3 = {}^5 P_3$

(ii)

Select '3' out of '5' & Arrange '3' persons on 3 seats. = No. of ways

$\hookrightarrow {}^n C_r = {}^5 C_3$



$= 6 = 3! \hookleftarrow$

$= {}^5 C_3 \times 3!$

$$\begin{aligned} {}^5 P_3 &= ({}^5 C_3) \times 3! \\ \Rightarrow \frac{5!}{(5-3)! 3!} &= {}^5 C_3 \end{aligned}$$

Properties of $n C_r$

★ ① $n C_r = \frac{n!}{(n-r)! r!}$ ★

$$n C_r = \frac{n P_r}{r!} \Rightarrow n P_r = n C_r \cdot r!$$

n $\nearrow$ upper suffix.
 $(r$ $\rightarrow$ Combination (selection)
 $)$ $\searrow$ lower suffix.

② $n C_0 = 1$

$$n C_0 = \frac{n!}{(n-0)! 0!} = \frac{\cancel{n!}}{\cancel{n!} \times 1} = 1$$

$$n C_n = 1$$

$$n C_n = \frac{n!}{(n-n)! n!} = \frac{\cancel{n!}}{0! \times \cancel{n!}} = 1$$

$$n C_1 = n \quad \checkmark$$

$$n C_{n-1} = n \quad \checkmark$$

Properties of nCr

③ $nCr = nC_{n-r}$

e.g. ${}^{10}C_3 = {}^{10}C_{10-3}$

${}^{10}C_3 = {}^{10}C_7$ $3+7=10$

e.g. ${}^{12}C_4 = {}^{12}C_8$ $4+8=12$

LHS $= {}^nC_r = \frac{n!}{(n-r)! r!}$ ✓

RHS $= {}^nC_{n-r} = \frac{n!}{[n-(n-r)]! (n-r)!}$
 $= \frac{n!}{r! (n-r)!}$ ✓✓

④ If $nCa = nCb$ then $\begin{cases} \text{(i) } a=b \text{ or} \\ \text{(ii) } a+b=n \end{cases}$

e.g. ${}^nC_7 = {}^nC_8$
 $n=?$
 $7+8=n$
 $15=n$

Properties of nC_r

$$(5) \quad {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$${}^7C_3 + {}^7C_4 = {}^8C_4$$

$${}^nC_r = \frac{n!}{(n-r)! r!}$$

Proof: LHS = ${}^nC_r + {}^nC_{r-1}$

$$\text{LHS} = \frac{n!}{(n-r)! \cdot r!} + \frac{n!}{(n-r+1)! (r-1)!}$$

$$= \frac{n!}{(n-r)! \cdot r \cdot (r-1)!} + \frac{n!}{(n-r+1) \cdot (n-r)! (r-1)!}$$

$$= \frac{n!}{(n-r)! (r-1)!} \cdot \left\{ \frac{1}{r} + \frac{1}{n-r+1} \right\}$$

$$= \frac{n!}{(n-r)! (r-1)!} \cdot \left\{ \frac{n-r+1 + r}{r(n-r+1)} \right\}$$

$$= \frac{(n+1)!}{(n-r+1)! r!}$$

$$\text{RHS} = {}^{n+1}C_r$$

$$= \frac{(n+1)!}{(n+1-r)! r!}$$

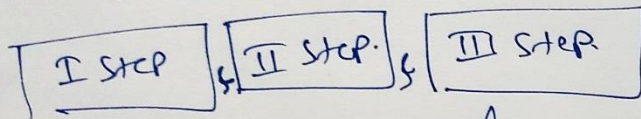
LHS = RHS

Fundamental Principle of Counting :

Before Example (19)

① Multiplication Rule (&)
(and)

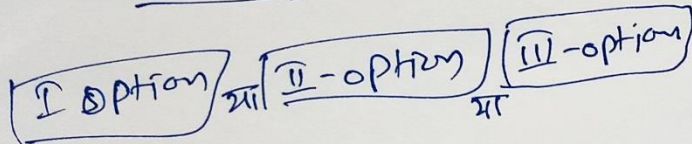
Project.



$$= m \times n \times l$$

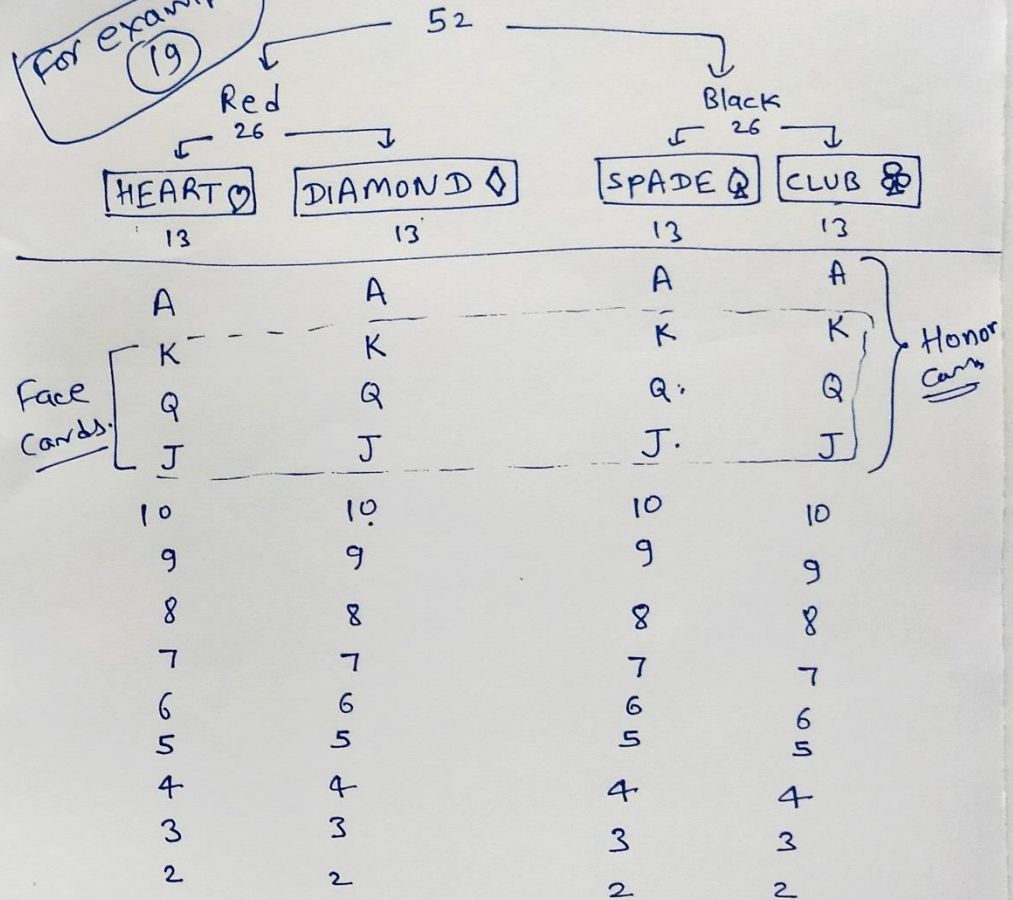
② Addition Rule (or)
(या)

Project.



$$m + n + l$$

For example (19)



Example - [19]

No. of ways of choosing

$$4 \text{ cards from } 52 \text{ playing cards} = {}^{52}C_4$$

(i) 4 cards are of the same suit = either Heart or Dia. or Spade or club

$$= {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4$$

(ii) 4 cards belong to the 4 different suits = 1 from Heart & 1 from Diamond & 1 from Spade & 1 from club

$$= {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1$$
$$= (13)^4$$

(iii) are face cards $\rightarrow 12 = \text{total}$.

$$= {}^{12}C_4 = \frac{12!}{8! 4!}$$

(iv) 2 are red & 2 are black

$$= {}^{26}C_2 \times {}^{26}C_2$$

$$= ({}^{26}C_2)^2$$

(v) Cards are of the same colour either Red Cards or Black

$$= {}^{26}C_4 + {}^{26}C_4$$

$$= 2 \times {}^{26}C_4$$

Revision:

$$(1) \quad nC_r = \frac{n!}{(n-r)! r!}$$

$$(2) \quad nC_r = nC_{n-r} \quad \rightarrow \quad {}^{13}C_2 = {}^{13}C_{11}$$

$$(3) \quad nC_a = nC_b \quad \begin{cases} \rightarrow (i) \quad a=b \text{ or} \\ \rightarrow (ii) \quad \underline{a+b=n} \star \end{cases}$$

Extra Prop.

$$\underline{\underline{nC_r + nC_{r-1} = {}^{n+1}C_r}}$$

Q.1

$$nC_8 = nC_2 ; \text{ Find } \underline{nC_2}$$

By property (3).

$$8 \neq 2$$

then

$$8+2=n$$

$$\Rightarrow \underline{n=10}$$

$$nC_2 = {}^{10}C_2$$

$$= \frac{10!}{8! 2!}$$

$$= \frac{5 \cancel{10} \times 9 \times \cancel{8!}}{\cancel{8!} \times 2 \times 1} = 45 \quad \checkmark$$

Q.2 (i) ${}^{2n}C_3 : {}^nC_3 = 12 : 1$

★ ${}^nC_r = \frac{n!}{(n-r)! r!}$

$$\Rightarrow \frac{\left[\frac{(2n)!}{(2n-3)! \cancel{3!}} \right]}{\left[\frac{n!}{(n-3)! \cancel{3!}} \right]} = \frac{12}{1}$$

$$\Rightarrow \frac{\frac{(2n)(2n-1)(2n-2)(2n-3)!}{(2n-3)!}}{\frac{n \cdot (n-1)(n-2) \cdot (n-3)!}{(n-3)!}} = \frac{12}{1}$$

$$\Rightarrow \frac{4(2n-1)}{n-2} = \frac{12}{1}$$

$$\Rightarrow 2n-1 = 3n-6$$

$$\Rightarrow 6-1 = 3n-2n \Rightarrow \boxed{5=n}$$

(ii) ${}^{2n}C_3 : {}^nC_3 = 11 : 1$

$$\Rightarrow \frac{4(2n-1)}{n-2} = \frac{11}{1}$$

$$\Rightarrow 8n-4 = 11n-22$$

$$\Rightarrow 22-4 = 11n-8n$$

$$\Rightarrow 8n = 18$$

$$\boxed{n=6} \checkmark$$

Revision, ① n_{C_r} = No. of combinations (selection)
of r -objects taken from n -objects.

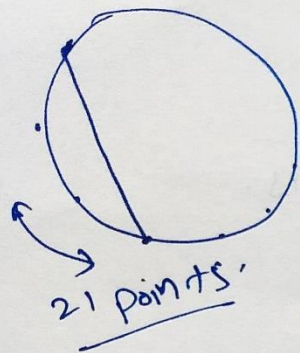
$$\textcircled{2} \quad n_{C_r} = \frac{n!}{(n-r)! r!} \quad (0 \leq r \leq n)$$

③ multiplication Rule ($\rightarrow$ &) $\overbrace{m}^{\text{I Step}} \times \overbrace{n}^{\text{II Step}} = m \cdot n$

Addition Rule (or) $\underbrace{m}_{\text{I option}} + \underbrace{n}_{\text{II-option}} = m+n$

Q.3

21 points on a ~~circle~~ Circle



→ one chord is made by joining two particular points of circle.

→ So we have to select 2 points to make a chord.
↓
(out of 21 points.)

No. of chords = No. of ways to select 2 points of the circle (out of 21 points.)

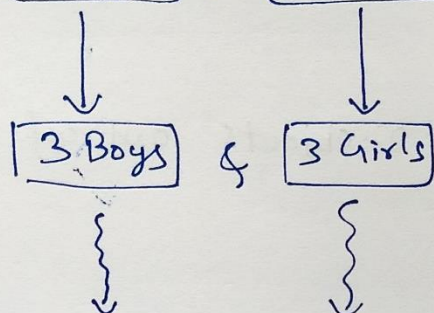
$$= {}^{21}C_2$$

$$= \frac{21!}{19! \cdot 2!} = \frac{21 \times \overset{10}{\cancel{20}} \times \cancel{19!}}{\cancel{19!} \times \cancel{2} \times 1} = 210 \checkmark$$

Q.4

from 5 Boys & 4 Girls

Select a team



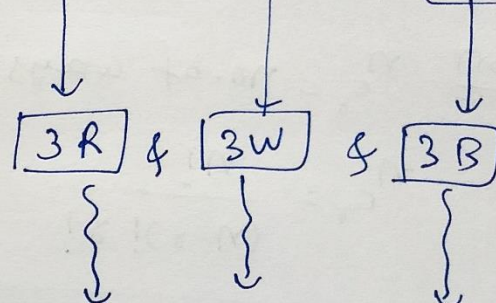
No. of ways to select a team

$$\begin{aligned} &= {}^5C_3 \times {}^4C_3 \\ &= \frac{5!}{2!3!} \times \frac{4!}{1!3!} \\ &= \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 3 \cdot 2 \cdot 1} \times \frac{4 \cdot 3 \cdot 2 \cdot 1}{1 \cdot 3 \cdot 2 \cdot 1} \\ &= 5 \times 2 \times 4 \\ &= 40 \end{aligned}$$

Q.5

from 6 Red, 5 white, 5 Blue

Select total 9-Balls



No. of ways to select 9 balls

$$\begin{aligned} &= {}^6C_3 \times {}^5C_3 \times {}^5C_3 \\ &= \frac{6!}{3!3!} \times \frac{5!}{2!3!} \times \frac{5!}{2!3!} \\ &= \frac{6 \cdot 5 \cdot 4 \cdot 3!}{3! \cdot 3!} \times 10 \times 10 \\ &= 20 \times 10 \times 10 \\ &= 2000 \end{aligned}$$

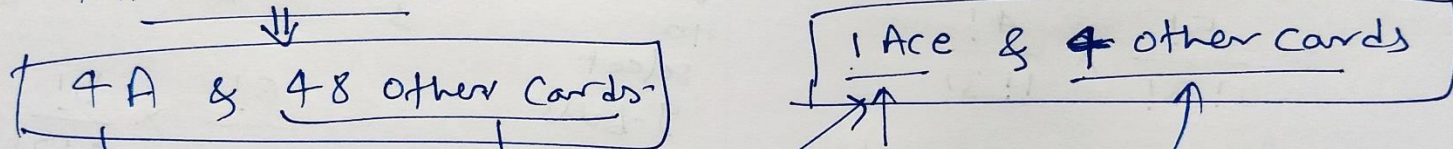
(if 3 balls of each color)

Revision $nCr =$ no. of ways of selecting r -objects out of n -objects.

$$nCr = \frac{n!}{(n-r)! r!}$$

Q.6

Total 52 Cards $\xrightarrow{\text{Select}}$ 5 cards.



Total No.
of ways

$$= \binom{4}{1} \times$$

$$\binom{48}{4}$$

$$= 4C_1 \times 48C_4$$

$$= 4 \times 48C_4$$

Q.7

from
total 17
players

5 Bowlers

12 others

← Available

Select Cricket
team of 11
players

4 Bowlers
Exactly

7 others

← Desire.

Total No. of
ways of

Selecting a Cricket
team of 11-players

$$= {}^5C_4 \times {}^{12}C_7 = {}^5C_4 \cdot {}^{12}C_7$$

$$= \frac{5!}{1!4!} \cdot \frac{12!}{\cancel{5!}7!}$$

$$= \frac{12 \cdot 11 \cdot 10 \cdot \cancel{9} \cdot \cancel{8} \cdot \cancel{7}!}{\cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1 \cdot \cancel{7}!}$$

$$= 12 \times 11 \times 10 \times 3$$

$$= 3960 \checkmark$$

8

Bag

5 Black
Balls

6 Red
Balls

No. of ways of selecting 2 Black & 3 Red
Balls

$$= {}^5C_2 \times {}^6C_3$$

$$= \frac{5!}{3!2!} \times \frac{6!}{3!3!}$$

$$= \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 2 \cdot 1} \times \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 3 \cdot 2 \cdot 1}$$

$$= 10 \times 20 = 200$$

9

Compulsory

2 specific → Fixed by
Question itself.

Available
9 Courses

2 Specific
Compulsory

+ 7 others

Choose
5 Courses

2 Specific
Compulsory

& 3 others
optional

$${}^2C_2 = 1$$

No. of
ways

$$= 1 \times {}^7C_3$$

of choosing

the programme

$$= 1 \times {}^7C_3$$

$$= 1 \times \frac{7!}{4!3!}$$

$$= \frac{7 \cdot 6 \cdot 5 \cdot 4!}{4! \cdot 3!}$$

$$= 35$$

Q.1 'DAUGHTER' Available

Vowels → Consonants

③ A, U, E

⑤ D, G, H, T, R

$$\frac{n!}{(n-r)! r!}$$

$nCr = \text{Selection}$

Combination
(Selection)

Desired

2 vowels

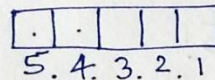
$$\downarrow 3C_2$$

3 consonants.

$$\downarrow 5C_3$$

Repetition
↓
No.

Different words



$$= 5! = 120$$

(Arrangement)

Permutation
(Arrangement)

Total No. of words

= No. of ways of Selection (Combination) × No. of ways of Arrangement (Permutation)

$$= 3C_2 \times 5C_3 \times 120$$

$$= \frac{3!}{1! 2!} \times \frac{5!}{2! 3!} \times 120 = \frac{3 \times 2 \times 1}{1 \times 2 \times 1} \times \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1} \times 120 = 3600$$

Q.2 'EQUATION'

All the letters

Repetition
↓
Not Allowed

Vowels Together

'EUAIO'

Bag_v

AEIOU

Consonants together

'QTN'

Bag_c

QTN

For e.g.

AEIOU QTN → ✓

EIAUO INQ → ✓

TNQ EIAUO ✓

Arrangement of Bags = 2

V C
C V

Arrangement of vowels in their own Bag = 5!

AEIOU

1 1 1 1 1
5 4 3 2 1

Arrangement of consonants in their own Bag = 3!

QTN

3 2 1
1 1 1

Total no. of words = $2 \times 5! \times 3!$

$$= 2 \times 120 \times 6 = 1440 \checkmark$$

$${}^nC_r = \frac{n!}{(n-r)! r!}$$

Q.3 Total Available = 9 Boys & 4 Girls
Form Committee by 7 members

(i) Exactly 3 Girls

Committee 7-member

& $\begin{array}{|c|} \hline 3G \\ \hline 4B \\ \hline \end{array} \rightarrow \text{selection } {}^4C_3$
 $\begin{array}{|c|} \hline 3G \\ \hline 4B \\ \hline \end{array} \rightarrow {}^9C_4$

$$= {}^4C_3 \times {}^9C_4$$

(Girls) (Boys)

$$= \frac{4!}{1!3!} \times \frac{9!}{5!4!}$$

$$= \frac{4 \cdot 3!}{1 \times 3!} \times \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5!}{5! \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

$$= 4 \times \frac{9 \cdot 7 \cdot 2}{126}$$

$$= 504$$

(ii) atleast 3 Girls
(अन्य से कम) $\rightarrow 3, 4G$

$\begin{array}{|c|} \hline 3G \\ \hline 4B \\ \hline \end{array}$ or $\begin{array}{|c|} \hline 4G \\ \hline 3B \\ \hline \end{array}$

$$= {}^4C_3 \times {}^9C_4 + {}^4C_4 \times {}^9C_3$$

$$= \frac{4!}{1!3!} \times \frac{9!}{3!4!} + \frac{4!}{0!4!} \times \frac{9!}{6!3!}$$

Answer.

(iii) atmost 3 girls

(ज्यादा से ज्यादा)

$\begin{array}{|c|} \hline 3G \\ \hline 4B \\ \hline \end{array}$ or $\begin{array}{|c|} \hline 2G \\ \hline 5B \\ \hline \end{array}$ or $\begin{array}{|c|} \hline 1G \\ \hline 6B \\ \hline \end{array}$ or $\begin{array}{|c|} \hline 0G \\ \hline 7B \\ \hline \end{array}$

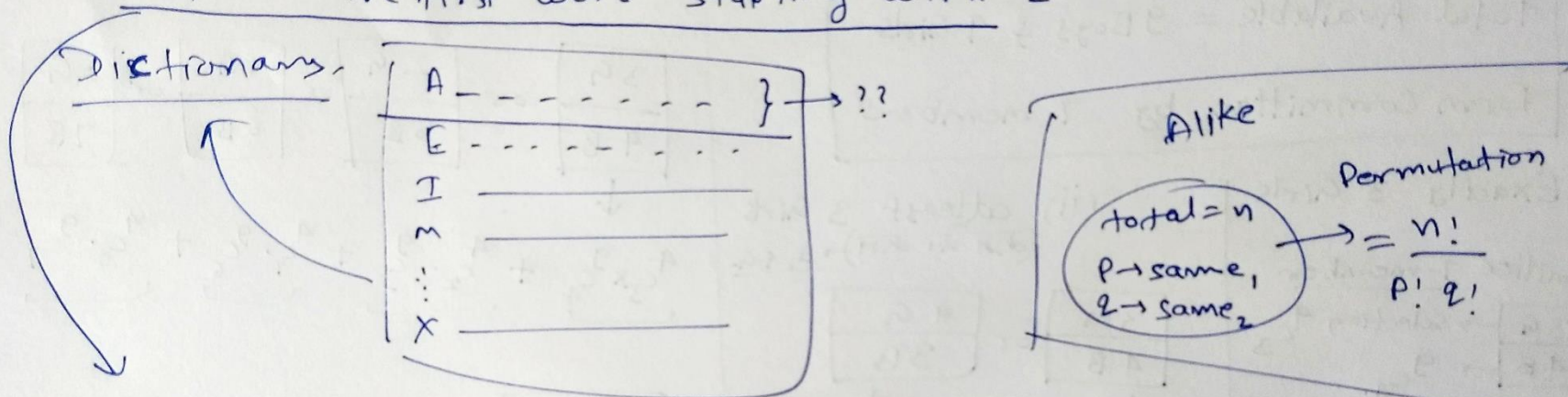
$$= {}^4C_3 \times {}^9C_4 + {}^4C_2 \times {}^9C_5 + {}^4C_1 \times {}^9C_6 + {}^4C_0 \times {}^9C_7$$

Answer

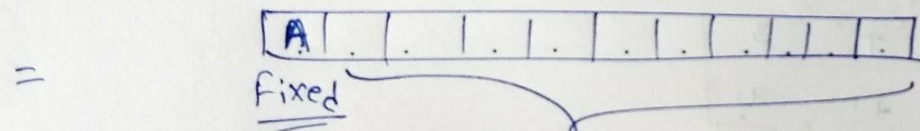
Q.4 'EXAMINATION'

In alphabetical order $\rightarrow$ AA E II M NN O T X $\rightarrow$ total letters = 11

Before the first word starting with 'E'

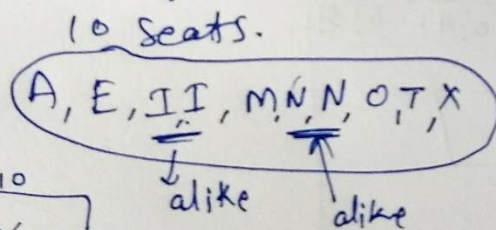


No. of words starting ~~the~~ with the letter 'A'



$$= \frac{10!}{2! \cdot 2!}$$

$$= \frac{10 \cdot \overbrace{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}^{10} \cdot \overbrace{4 \cdot 3 \cdot 2 \cdot 1}^{10}}{2 \times 2}$$



Restriction
Priority

$$\begin{array}{r} 126 \\ \times 72 \\ \hline 252 \\ 882 \times \\ \hline 9072 \end{array}$$

$$= 10 \times 10 \times \underline{72} \times \overbrace{42}^{126} \times 3 = 907200$$

5. Available = 0, 1, 3, 5, 7, 9

6-Digit No.

"Divisible" by 10

Fixed, 0

Priority

Restriction (Priority)

5 · 4 · 3 · 2 · 1

$= 5 \times 4 \times 3 \times 2 \times 1$

$= 5!$

(Last Digit = 0)

No. of ways. = 120

6

Available

5 Vowels

21 Consonants

Select

Desired

2 Vowels

2 Consonants

only selection

$5C_2 \times 21C_2$

Arrange

A E R T -
R A T E :-

After Selection Available $\rightarrow 4$ letters

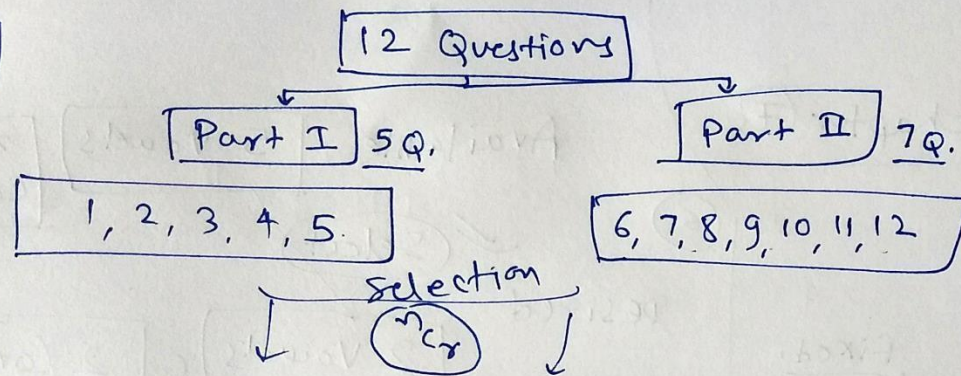
$(2V, 2C)$

4 · 3 · 2 · 1

Total No. of words = $\frac{4 \times 3 \times 2 \times 1}{\text{Arrange}} \times \frac{5C_2 \cdot 21C_2}{\text{selection}}$

$= 4 \cdot 3 \cdot 2 \cdot 1 \cdot \frac{5!}{3!2!} \times \frac{21!}{19!2!}$

Q. 7



A Hmpt
8
Questions

at least 3
from each Part

Case-I.

or

Case-II.

or

Case-III.

3 Q (I) & 5 Q (II)

4 Q (I) & 4 Q (II)

5 Q (I) & 3 Q (II)

$$= \left[{}^5C_3 \times {}^7C_5 \right]$$

$$+ \left[{}^5C_4 \times {}^7C_4 \right]$$

$$+ \left[{}^5C_5 \times {}^7C_3 \right]$$

= Answer

$$nCr = \frac{n!}{(n-r)! r!}$$

✓

Q.8

52 Cards Total $\xrightarrow{\text{Select}}$ 5 Cards

4 Kings & 48 Others

1 King & 4 other Cards

Select.

Exactly

No. of
ways

$$= {}^4C_1$$

X

$$48C_4$$

= ~~Ans~~ Answer.

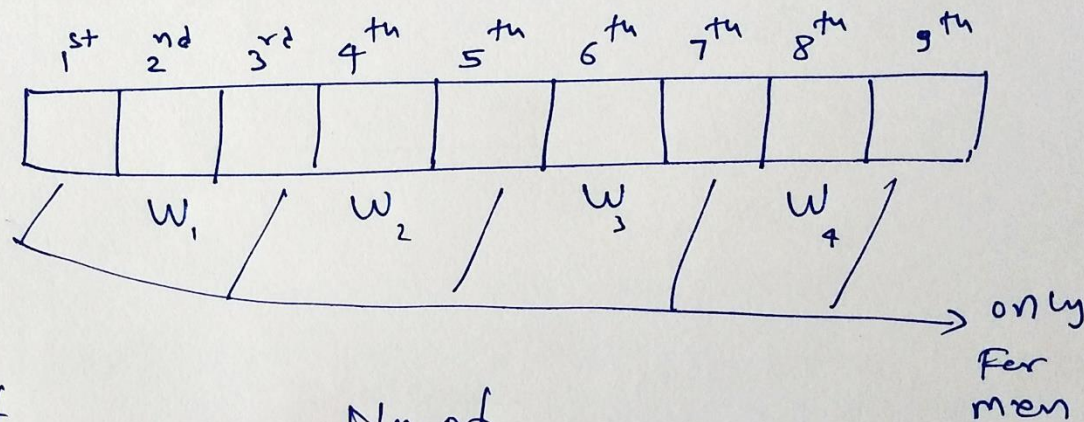
★

$${}^nC_r = \frac{n!}{(n-r)! r!}$$

[Q.9] Available $\rightarrow$ 5 men, 4 women

Women occupy Even places

Total seats = 9



$$\begin{aligned} \text{No. of ways} &= \text{No. of Arrangements of women} \times \text{No. of Arrangements of men} \\ &= \begin{matrix} \square_{2^{\text{nd}}} & \square_{4^{\text{th}}} & \square_{6^{\text{th}}} & \square_{8^{\text{th}}} \end{matrix} \times \begin{matrix} \square_{1^{\text{st}}} & \square_{3^{\text{rd}}} & \square_{5^{\text{th}}} & \square_{7^{\text{th}}} & \square_{9^{\text{th}}} \end{matrix} \\ &= (4 \times 3 \times 2 \times 1) \times (5 \times 4 \times 3 \times 2 \times 1) \\ &= 4! \times 5! \\ &= 24 \times 120 \\ &= 2880 \end{aligned}$$

Q.10

25 Students

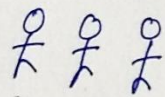


Special '3'

22 others

Party 10 Students

Case I



Others

Either all of them

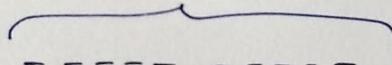
$$= {}^3C_3 \times {}^{22}C_7$$

$= 1 \times {}^{22}C_7$

+

Case II

all 10 other



or none of them

$$= {}^{22}C_{10}$$

+

$$nCr = \frac{n!}{(n-r)! r!}$$

$$\begin{aligned} \text{Total No. of ways} &= {}^{22}C_7 + {}^{22}C_{10} \\ &= \frac{22!}{15! 7!} + \frac{22!}{12! 10!} \end{aligned}$$

Ans.

Q.11 ASSASSINATION $\rightarrow$ AAA, II, NN, O, T, SSSS,

All the S's are
together

Bag method,

AAA, II, NN, O, T, SSSS

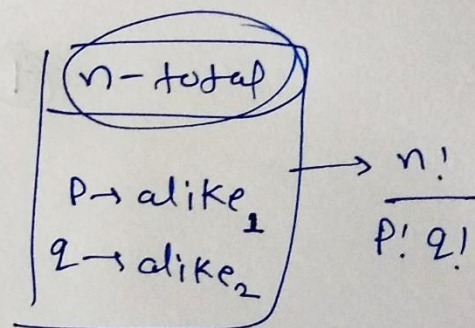
10 objects $\rightarrow$ Permutations
No. of words = $\frac{10!}{3! 2! 2!}$

AAA II NN

$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1}$$

$$= 10 \times 10 \times 72 \times 21$$

$$= 151200$$



$$\begin{array}{r} 72 \\ \times 21 \\ \hline 144 \\ \times 1512 \\ \hline \end{array}$$